

a chewing gum stick, was developed as a hobby project but has become increasingly popular in commercial applications like power management metering, medical devices, security devices, unmanned aerial vehicles, and wireless and handheld products. It has two product lines, one based on TI OMAP and the other based on Marvell XScale processor. Some of the other open source options are Leopard board and Maple board that allow full computing capability.

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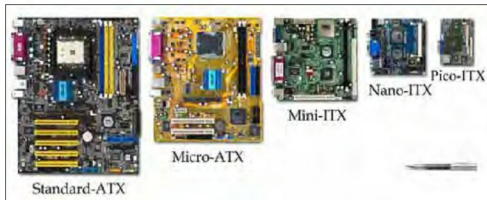
SBCs are traditionally known for industrial computing applications. However, Samsung recently introduced an SBC virtualisation environment to bring it to office. The environment will strengthen internal security and is based on a test project with Samsung SDS.

A company called Intercept Concept claims that its ultra-low-power single-board computer based on Freescale's PowerPC provides Gigabit Ethernet interfaces for high-performance network connectivity, redundant fail-safe links, powerful control elements for network switches, data storage sub-systems, network appliances, and print and imaging devices.

The latest offering from GE Intelligent Platforms, a 6U VMEbus SBC based on Freescale's QorIQ dual-core processor, delivers around ten times the processing performance depending on applications. Many such PowerXtreme SBCs are deployed in demanding military/aerospace applications in the US and Europe.

California-based ACCESS I/O Products offers multiple-function single boards through optically isolated inputs. This allows individual channel-to-channel isolation, that is, every channel is separated physically and electrically. The solid-state FET outputs capable of switching up to 3A each and two 16-bit analogue inputs make this module suitable for control and instrumentation application.

ADLINK Technology announced



The ITX form factors from VIA Technologies

its latest LittleBoard 735 EBX SBC based on Intel Atom N270 processor at 1.6 GHz, codenamed 'Diamondville,' at the ESC San Jose in April. The board combines Navy Pier platform, Atom N270 with ADLINK Extreme Rugged design methodology to create a feature-rich upgrade path for previous end-of-life products. It is designed to tolerate shock, vibration, humidity and temperature extremes.

The FeaturePak Initiative, originated by Diamond Systems, was launched at the Embedded World tradeshow in March 2010. Gurunatham G.V., managing director, Electro Systems Associates, shares, "This new standard defines a highly-compact, low-profile and inexpensive way to add configurable input/output functions to embedded systems." He adds that the tiny FeaturePak modules that are just $4.3 \times 6.5 \text{ cm}^2$ ($1.70 \times 2.55 \text{ inch}^2$) in size can be used for providing snap-in options or upgrades for off-the-shelf single-board computers and computer-on-module baseboards, or as building blocks to facilitate the development of fully-custom embedded electronics.

Gaming is one of the most aggressively growing industries. "Single-board computers of MB 62000 series are designed specifically for the gaming and amusement industry. These are based on Intel's GMA 900 graphics engine for 2-D/3-D applications and are supported by VGA+LVDS dual-display option. They also feature high-level security by TPM 1.2, FPGA and Authenix chip, and are ideally meant for video slot machines, amusement

game machines, master units of roulette machines, downloadable gaming terminals, high-end vending machines, ATM/cash machines and kiosks. Gaming manufacturers benefit from single-board features like integrated and economical assembly and purchasing costs," shares Rajan.

In sync with technological advancements

The concept of single-board computer is now new. It's a system that has survived for more than two decades now and evolved with changing technologies. By principle, it is a single board, but expandability is the most important feature of single-board computers. Due to standardisation and aggregation, expansion options are backward-compatible and forward-looking.

With the advancement in silicon technologies, high-end processors like ARM-based processors and Intel's multi-core processors have found a space on the tiny board to support high-performance products.

Connectivity is another trend that has been well-incorporated in single-board computers. SBCs are ready with Wi-Fi, 3G, Ethernet, USB, Bluetooth, ZigBee and other connectivity options. As embedded products become more interactive, display options are supported on single-board computers to make these a complete, feature-rich, ready-to-use solution for a variety of applications. ●

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